THE REMIX

Triple Differences

Day 2 of 3, Morning: When parallel trends fails, find a third group

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Yesterday's closing claim, today's tension

> *DiD is a research design with two ingredients — the 2$\times$2 calculation and parallel trends — and together they identify the ATT.*

— Day 1 closing slide

- Yesterday: **if you believe PT, the 2$\times$2 identifies the ATT.**
- Today: **what if you don't believe PT?**
- Two answers — both responses to a PT violation. This morning: **a third group**. This afternoon: **covariates**.

Some US states required firms to cover maternity benefits

In the 1970s.

Same terms as any other medical condition.

Other states did not.

Did the law lower women's wages?

The question was about **incidence**.

Did the cost of the mandate fall on
married women aged 20–40?

The obvious move is a 2×2

$$\widehat{\delta}_{2\times 2} = \left(\overline{Y}^{\text{mandate}}_{\text{post}} - \overline{Y}^{\text{mandate}}_{\text{pre}}\right) - \left(\overline{Y}^{\text{control}}_{\text{post}} - \overline{Y}^{\text{control}}_{\text{pre}}\right)$$

Four cell means, three subtractions. Day 1, all over again.

But parallel trends might not hold

$$(NJ_t - PA_t) \neq 0$$

The two state groups look too different.

The event study you have on hand isn't flat in the pre.

What do you do instead?

In 1994, Gruber introduced a new design

Gruber, J. (1994). *The Incidence of Mandated Maternity Benefits.* *American Economic Review* 84(3): 622–641.

Based on his **Harvard dissertation** (1992).

Advised by Larry Summers. Larry Katz on the committee.

The Section keeps showing up

Orley Ashenfelter $\longrightarrow$ Henry Farber $\longrightarrow$ Larry Katz $\longrightarrow$ Jonathan Gruber

*Katz has chaired or served on **250+** dissertation committees.*
The credibility revolution has many roots. Princeton's IRS is one of them, again.

This morning, six beats

1. Gruber 1994: the canonical DDD application
2. Two biased DiDs — and one parallel-bias assumption
3. The 8-cell calculation, by hand
4. A simulation: see the bias, see DDD recover the ATT
5. DDD as a regression specification
6. When DDD goes wrong — and Gruber's actual numbers

LESSON

1. Gruber and the Maternity Mandate

The policy: 1976 maternity insurance mandates

- Several states in 1976 mandated that employer-provided health insurance cover *childbirth on the same terms as any other medical condition*.
- Before: pregnancy could be excluded from coverage.
- Cost to insurers $\rightarrow$ cost to employers $\rightarrow$ **cost to whom?**

The economic question

Did the burden shift back to childbearing-age women in the form of **lower wages**? Or did employers absorb it? Or was it shifted to consumers?

Whose wages would have to fall?

If the mandate truly raises the employment cost of childbearing-age women, classical incidence reasoning predicts the cost shifts to *them* via lower wages.

- Treated cell: **married women 20–40** in mandate states.
- Comparison group — naive: married women 20–40 in non-mandate states.
- But: women's labor-market outcomes were diverging across states for many reasons (manufacturing decline, services growth, two-earner-household norms). **PT is suspect.**

Gruber’s move: bring in a third group

- In the same state, in the same labor market, take a group the mandate *should not touch*: **single men and older women.**
- They live in the same economy. The same trends hit them.
- But they’re not the ones the mandate raised the cost of employing.

The bet

If the differential state trend hits the treated group AND the placebo group equally hard, the placebo group’s “DiD” just *is* the bias.

Subtract the placebo DiD from the naive DiD — get the ATT.

LESSON

2. Two Biased DiDs, One Bias

Biased DiD #1 — married women, two states

	Mandate states (NJ)	Control states (PA)
Pre	$Y = NJ$	$Y = PA$
Post	$Y = NJ + NJ_t + D$	$Y = PA + PA_t$
Difference (post-pre)	$NJ_t + D$	PA_t

$$\widehat{\delta}_{\text{DiD}} = (NJ_t + D) - PA_t = D + \underbrace{(NJ_t - PA_t)}_{\text{bias if } NJ_t \neq PA_t}$$

The naive DiD recovers the ATT *only if* the state trends were the same. **Assume they aren't.**

Biased DiD #2 — single men + older women, same two states

	Mandate states (NJ)	Control states (PA)
Pre	$Y = NJ$	$Y = PA$
Post	$Y = NJ + NJ_t$	$Y = PA + PA_t$
Difference (post-pre)	NJ_t	PA_t

$$\widehat{\delta}^{\text{placebo}}_{\text{DiD}} = NJ_t - PA_t = NJ_t - PA_t$$

No treatment here. The placebo DiD *is* the bias — nothing else.

Subtract the bias — triple difference

$$\widehat{\delta}_{\text{DDD}} = \widehat{\delta}_{\text{DiD}} - \widehat{\delta}^{\text{placebo}}_{\text{DiD}}$$

$$= [D + (NJ_t - PA_t)] - (NJ_t - PA_t) = D$$

The new identifying assumption: parallel bias

$$(NJ_t - PA_t)^{\text{married}} = (NJ_t - PA_t)^{\text{placebo}}$$

- **Not** "two parallel trends." One assumption: the *differential* state trend hits both groups the same.
- Trends can differ. Levels can differ. The cross-state *gap* in trends has to be the same across the two demographic groups.

The formal version — Olden & Møen (2022)

Same idea, in potential-outcomes form:

$$\underbrace{E[\Delta Y^0 \mid D=1, G=1] - E[\Delta Y^0 \mid D=0, G=1]}_{\text{bias in the main DiD (eligible group)}}$$

$$=$$

$$\underbrace{E[\Delta Y^0 \mid D=1, G=0] - E[\Delta Y^0 \mid D=0, G=0]}_{\text{bias in the placebo DiD (ineligible group)}}$$

- D: treatment state. G: eligible group. ΔY^0: change in the *untreated* potential outcome.
- Each component DiD has its own non-parallel-trends bias.
- DDD identifies the ATT iff the two biases are **equal** — not zero.

Olden & Møen (2022), Econometrics Journal 25(3). They call it the bias stability assumption (after Frölich & Sperlich 2019).

What I wrote in the *Mixtape*, 1st edition

> *The effect can be isolated if the gap between high- and low-wage employment would have evolved similarly in the treatment state counterfactual as it did in the historical control states.*

— *Causal Inference: The Mixtape*, 1st ed.

Right idea — verbally. Olden & Møen (2022) wrote it in expectations.

Method papers fill quiet gaps

DDD has been used in top journals since **Gruber 1994**.

For nearly three decades, nobody wrote down its identifying assumption in expectations.

The literature ran on intuition.

Read the method papers. The 2nd edition (Yale UP, 2026) now carries Olden & Møen's equation.

DDD is a design, not a falsification

- A **design** has an identifying assumption. The assumption is what delivers the estimate.
- A **falsification** is a test or a conjecture — a probe of whether some *other* design's assumption is credible.

Same procedure, different jobs

Placebo DiD as falsification: run DiD on a group whose outcome shouldn't move. If it doesn't, your original DiD's parallel trend looks credible. That's a *check on DiD*.

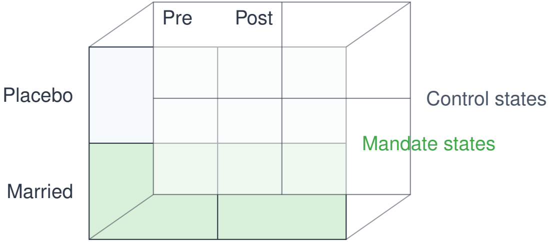
Placebo DiD as DDD: subtract the placebo DiD from the main DiD. The difference *is* the estimate. That's a *new design*, with its own assumption (parallel bias).

The arithmetic looks identical. The interpretation is not. Identifying assumption $\Rightarrow$ design; conjecture $\Rightarrow$ falsification.

LESSON

3. The 8 Cells

The DDD cube — two states × two periods × two groups



8 cells: states × groups × periods

We need 8 averages. We will form 3 subtractions twice over — a 2×2 DiD for each group — then subtract the placebo DiD from the treated DiD. One number.

The 8 cells, filled in from the simulation

	Mandate states		Control states	
	Pre	Post	Pre	Post
Married women 20–40	70,503	73,813	49,006	60,013
Placebo (men + older women)	71,510	82,490	50,003	63,742

- **Naive DiD** (married women): $(\$57{,}488 - \$58{,}722) - (\$70{,}224 - \$66{,}760) \approx -\$7{,}697$
- **Placebo DiD** (men + older women): $\approx -\$2{,}759$
- **DDD**: $-\$7{,}697 \ - \ (-\$2{,}759) \ = \ -\$\mathbf{4{,}938}$

True ATT in the simulation: $-\$5{,}000$. DDD recovers it. Naive DiD is off by 54%.

Simulation: `zurich/labs/Triple-Diff/ddd_simulation.R`. 40 states $\times$ 50 cities $\times$ 3 worker groups $\times$ 11 years; seed 20200403.

The DDD formula — one line, eight cells

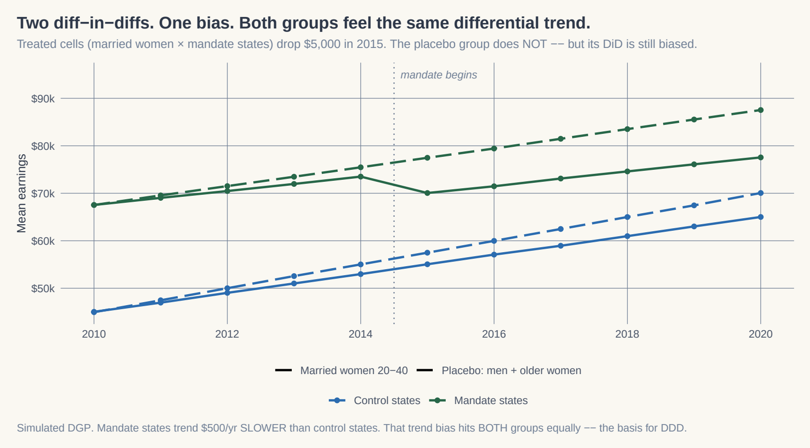
$$\widehat{\delta}_{\text{DDD}} = \left[(\overline{Y}_1^{TX,M} - \overline{Y}_0^{TX,M}) - (\overline{Y}_1^{AR,M} - \overline{Y}_0^{AR,M})\right]$$
$$- \left[(\overline{Y}_1^{TX,P} - \overline{Y}_0^{TX,P}) - (\overline{Y}_1^{AR,P} - \overline{Y}_0^{AR,P})\right]$$

- Superscripts: state (TX = mandate, AR = control), group (M = married, P = placebo)
- Subscripts: 0 = pre, 1 = post
- Read it as: **(DiD for the treated group)** minus **(DiD for the placebo group)**.
- Two DiDs. One subtraction. That is all.

Lesson

4. A Simulation Makes It Visible

See the bias before you read the regression



Mandate states (green) trend slower than control states (blue) — for BOTH groups.
That's the bias. The placebo group is in the same economy — it feels the same shift.

The simulation, code first

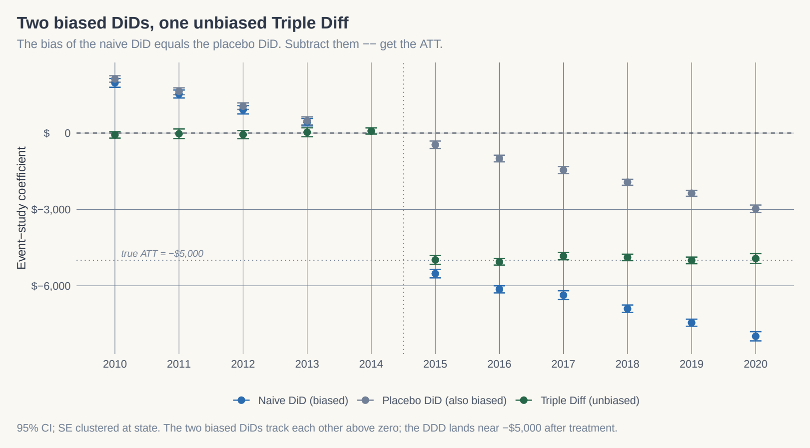
```
# DGP knobs:
# 40 states (1-20 mandate, 21-40 control)
# 3 worker groups (men, married women 20-40, older women)
# 2010-2020; treatment turns on in 2015
# TRUE ATT for married x mandate x post = -$5,000

df$state_trend <- if_else(df$experimental==1, 1000, 1500) # !! differential
df$group_trend <- if_else(df$worker==2, 500, 1000)
df$y0 <- df$baseline +
        df$state_trend * df$year_diff +
        df$group_trend * df$year_diff + rnorm(N, 0, 1500)

df$treat <- (df$experimental==1 & df$worker==2 & df$after==1)
df$y1 <- df$y0 - 5000 * df$treat
df$earnings <- df$treat * df$y1 + (1-df$treat) * df$y0
```

Full script: `decks/code/Triple-Diff/ddd_simulation.R`

Three event studies, one figure



Naive DiD (blue) lands too negative. Placebo DiD (gray) is the bias — nonzero even though nothing happened.

Triple Diff (green) lands at the true ATT of $-\$5{,}000$***.***

LESSON

5. DDD as a Regression

The saturated triple-interaction regression

$$\begin{aligned} Y_{ijt} &= \alpha + \beta_1 D_i + \beta_2 G_j + \beta_3 \tau_t + \beta_4 (D_i \times G_j) + \beta_5 (D_i \times \tau_t) \\ &\quad + \beta_6 (G_j \times \tau_t) + \boldsymbol{\delta}\,(D_i \times G_j \times \tau_t) + \varepsilon_{ijt} \end{aligned}$$

- D_i: state is a mandate state. G_j: worker is in the treated group (married women 20–40). τ_t: post-period.
- $\boldsymbol{\delta}$ is the triple interaction — the DDD estimate of the ATT.
- Every two-way interaction is also there. The triple strips out everything but the answer.

What the regression assumes

- **Parallel bias**: the differential state-trend that biases the treated DiD biases the placebo DiD by *the same amount*.
- **No anticipation**: the treated group didn't shift behavior before $t = 2015$.
- **SUTVA**: one unit's treatment doesn't change another unit's $Y(0)$.

What DDD does *not* require

Two parallel-trends assumptions (one per group). That misconception — DDD as a "double check" — is wrong.

DDD has *one* identifying assumption, and it's weaker than the DiD's. That's its appeal.

Lesson

6. Caveats, Gruber’s Actual Numbers, and When DDD Goes Wrong

Gruber (1994): the actual numbers (log hourly wages)

Location / year	**Pre-law**	**Post-law**	**Difference**
A. Treatment: Married women 20–40			
Experimental states	1.547 (.012)	1.513 (.012)	-0.034 (.017)
Control states	1.369 (.010)	1.397 (.010)	$+0.028$ (.014)
DiD (A)			$\mathbf{-0.062}$ (.022)
B. Placebo: Over 40 + Single Males 20–40			
Experimental states	1.759 (.007)	1.748 (.007)	-0.011 (.010)
Control states	1.630 (.007)	1.627 (.007)	-0.003 (.010)
DiD (B, placebo)			$\mathbf{-0.008}$ (.014)
		DDD = (A) $-$ (B) $=$	$\mathbf{-0.054}$ (.026)

DDD $=$ True DiD $-$ Placebo DiD. Placebo DiD $= -0.008$ — essentially zero, so the original DD (-0.062) was already approximately unbiased. DDD and DD agree because there was no differential trend to subtract.

Source: Gruber (1994) AER. Reproduced in Cunningham, Mixtape §9.

What this means for when DDD "mattered"

- Gruber's naive DiD: -6.2%
  Gruber's triple diff: -5.4%
  Very close. *Was DDD necessary?*
- In this case, no. The placebo wage trend was essentially zero, so the DD was already approximately unbiased.
- But that's the wrong question.

The honest answer

The value of DDD is what it would have caught. The placebo DiD was a real opportunity to invalidate the design. It didn't. The design is more credible because it *could have* failed.

This is the Popper move from Day 1, in a different costume.

When DDD is worse than DiD

- **The placebo group is also treated** — by a different policy that hit at the same time. Now the placebo DiD reflects *that* policy, and subtracting it injects bias instead of removing it.
- **The bias is not common** — the differential state-trend hits the treated group but not the placebo group (or hits with a different magnitude). DDD identifies a parameter you didn't want.
- **Wrong placebo group** — one that experiences the policy through indirect channels. Subtracting it removes signal, not bias.

The first defense is the placebo-group event study. If the placebo group has its own discontinuity at the treatment date, the placebo isn't placebo.

Even DDD wants its event study

- DDD is a design, not a black box. You owe the reader the same five pieces from Day 1:
  - **Bite**: did the policy bind?
  - **Event study**: pre-trends on the DDD coefficient (we built one — show it)
  - **Falsification**: a placebo for the *placebo group* itself
  - **Main result**: the DDD point estimate
  - **Mechanism**: the channel from policy to outcome
- **The event study is the way you defend parallel bias.** Plot it. Show that pre-period DDD coefficients are flat. Then the post-period estimate is credible.

What we’ve done this morning

1. Gruber 1994: a maternity-benefit story that DiD couldn’t tell
2. Two biased DiDs $\rightarrow$ one bias $\rightarrow$ subtract
3. The 8 cells, by hand, with the simulation matching
4. A picture showing parallel bias in trends
5. DDD as the saturated triple-interaction regression
6. One identifying assumption, not two — and how it fails

Lab over lunch: abortion legalization & gonorrhea (3×2)

Goal. Take the DDD machinery and run it on data Scott actually published with.

- Cunningham & Cornwell (2013), in the lineage of Donohue & Levitt (2001). **Treated cohort:** Black females 15–19 (`bf15`). **Placebo cohort:** Black females 25–29 (`bf25`).
- Repeal states vs. Roe states; pre/post the affected birth cohort. Eight cells.
- **Part 1:** compute the 8 cell means and the DDD by hand (weighted by `totpop`).
- **Part 2:** recover the same number via `lnr ~ repeal#cohort_treated#post`, clustered at `fip`.
- **Part 3:** compare hand vs. regression; check what happens if you cluster wrong.
- **Part 4:** write a paragraph on what the placebo cohort buys you over Donohue & Levitt's straight DD.

Walkthrough: `zurich/labs/Triple-Diff/README.md`. Data: `abortion.dta` on Scott's GitHub.

This afternoon

- DDD’s escape hatch needs a **third group** that shares the bias.
- Often you don’t have one.
- Then the move is: **condition on covariates**. Inverse propensity weighting, outcome regression, doubly robust.
- And one false friend: *controlling for the propensity score in a regression*. That’s not what DR is. We’ll see why.

Triple diff trades parallel trends for **parallel bias.**

Back after lunch — covariates.